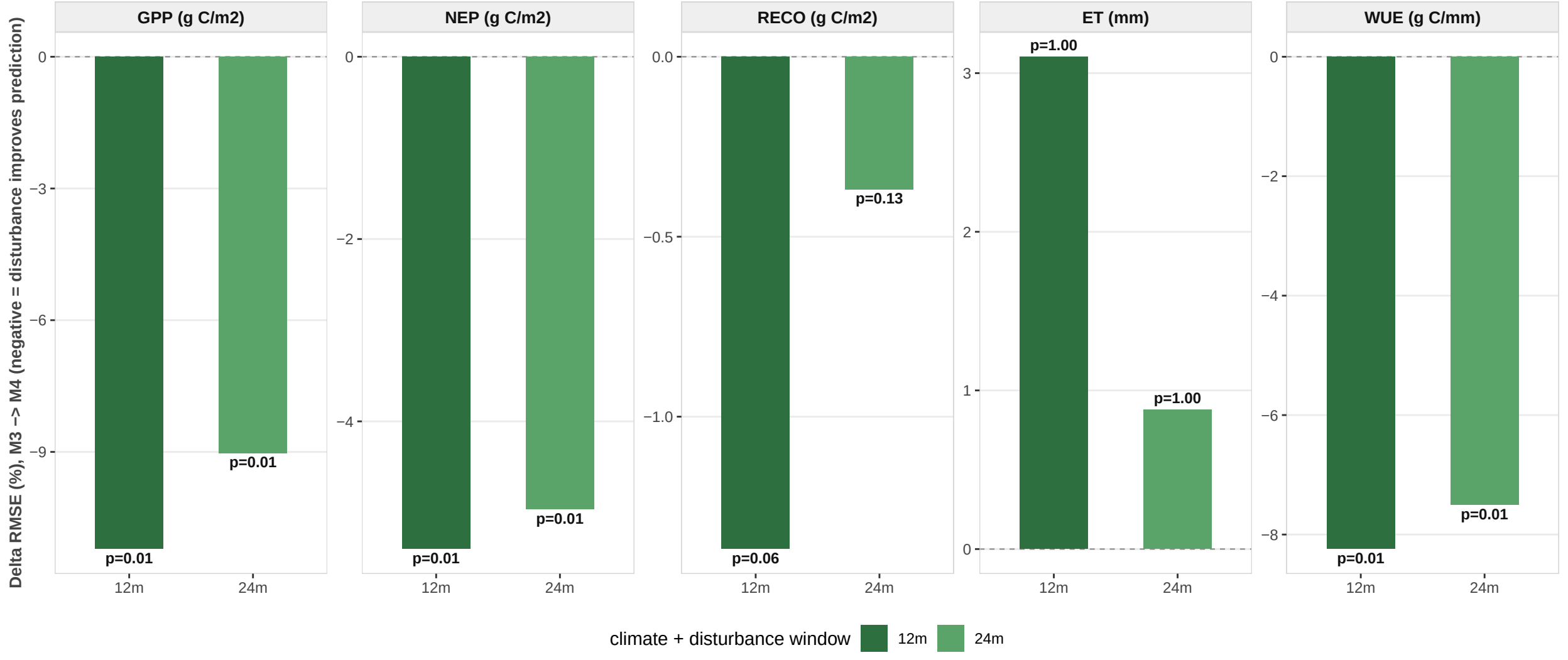
M3 −> M4 on flux totals: 12m vs 24m climate+disturbance window

tc>=30, 93 sites, n=378 site−years | 12m: climate=lag1, disturbance=contemporaneous+lag1 (100 cols) | 24m: +lag2 for both (110 disturbance cols)
XGBoost (GPPsat's tuned hyperparameters), LOSO CV | p from 99−rep block permutation of the disturbance block



12m shows consistently larger (or equal−direction) effects than 24m for every response. RECO moves from clearly null (24m, p=0.13) to borderline (12m, p=0.06) but neither clears p<0.05.
Same significance pattern otherwise: GPP/NEP/WUE real at both windows, ET null (and slightly positive, i.e. disturbance makes it marginally worse) at both.